

Meta-modeling soil organic carbon sequestration potential and its application at regional scale

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Abstract. Upscaling the results from process-based soil–plant models to assess regional soil organic carbon (SOC) change and sequestration potential is a great challenge due to the lack of detailed spatial information, particularly soil properties. Meta-modeling can be used to simplify and summarize process-based models and significantly reduce the demand for input data and thus could be easily applied on regional scales. We used the pre-validated Agricultural Production Systems sIMulator (APSIM) to simulate the impact of climate, soil, and management on SOC at 613 reference sites across Australia's cereal-growing regions under a continuous wheat system. We then developed a simple meta-model to link the APSIM-modeled SOC change to primary drivers, i.e., the amount of recalcitrant SOC, plant available water capacity of soil, soil pH, and solar radiation, temperature, and rainfall in the growing season. Based on high-resolution soil texture data and 8165 climate data points across the study area, we used the meta-model to assess SOC sequestration potential and the uncertainty associated with the variability of soil characteristics. The meta-model explained 74% of the variation of final SOC content as simulated by APSIM. Applying the meta-model to Australia's cereal-growing regions reveals regional patterns in SOC, with higher SOC stock in cool, wet regions. Overall, the potential SOC stock ranged from 21.14 to 152.71 Mg/ha with a mean of 52.18 Mg/ha. Variation of soil properties induced uncertainty ranging from 12% to 117% with higher uncertainty in warm, wet regions. In general, soils in Australia's cereal-growing regions under continuous wheat production were simulated as a sink of atmospheric carbon dioxide with a mean sequestration potential of 8.17 Mg/ha.

Key words: APSIM; Australia; carbon sequestration potential; cropland; crop model; meta-model; simulation; soil carbon; soil type; uncertainty.

INTRODUCTION

Agricultural soils may have substantial potential for sequestering carbon (C) through adopting conservation agriculture practices (CAPs), thereby contributing to climate change mitigation (Lal 2004, Smith 2004, Luo et al. 2010a). Lal (2004) estimated that the adoption of CAPs could potentially sequester 0.4–0.8 Pg C/yr in world cropland soils, which is 33.3–100% of the total potential C sequestration of world soils. For Australia's cereal-growing regions, Dalal and Chan (2001) estimated that CO₂ emissions would be reduced by >1.04 Pg after 20 years of adopting CAPs (nearly twice Australia's current total annual CO₂ emissions). However, the potential C sequestration capacity of the soil will be dependent on rates of primary productivity and soil organic matter decomposition regulated by climate, soil, and management (Luo et al. 2010a). The regional potential of soil organic carbon (SOC) sequestration

and particularly the uncertainty associated with the estimation of such potential in Australian croplands are rarely studied due to the limited information on soil, climate, and management conditions.

Process-based biogeochemical models such as APSIM (Wang et al. 2002, Keating et al. 2003), Century (Parton et al. 1987), DNDC (Li et al. 1994), and RothC (Jenkinson 1990), have been widely used to quantify the potential of C sequestration in agricultural soils at a field scale. Attempts have also been made to upscale these field scale results to assess C sequestration potential at regional or continental scales (Vleeshouwers and Verhagen 2002, Grace et al. 2006, Smith et al. 2007). However, these process-based models usually require detailed input information that is typically not available at regional and/or continental scales. This often leads to great uncertainty in simulation results and makes upscaling to regional scale a challenge (Ogle et al. 2006, 2010, Tang and Zhuang 2008, Goidts et al. 2009, Smith et al. 2010).

For a given location and time period, SOC change is the net balance between carbon input (primary produc-

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TABLE 1. A comparison of input requirements for Agricultural Production Systems sIMulator (APSIM) and the meta-model to predict soil organic carbon (SOC).

Inputs	APSIM	Meta-model
Management	crop cultivar; sowing date and depth; row spacing; fertilization; residue retention rate	optimal management was considered in this study in order to assess potential of SOC sequestration
Climate variables	daily maximum temperature; daily minimum temperature; daily precipitation; daily radiation	long-term mean temperature (T); long-term mean precipitation (P); long-term mean radiation (R)
Soil carbon and nitrogen-related inputs	for each soil layer: initial SOC content; initial size of microbial biomass pool; initial fraction of inert carbon; initial amount of fresh organic matter; initial nitrogen (nitrate and ammonia) contents; soil carbon to nitrogen ratio; soil pH	initial inert organic carbon in the 0–30 cm soil layer (IOC); soil pH
Soil water-related inputs	for each site: soil albedo; curve numbers for runoff calculation; soil evaporation parameters; for each soil layer: bulk density; air-dry water content; saturation water content; drained upper limit; –1.5 MPa lower limit; crop lower limit; initial soil water content	plant available water capacity of the soil profile (PAWC)

tivity and fraction of carbon returned) and output (carbon decomposition). At regional scales the SOC balance might be expressed as a simplified function of the underlying primary controls. Simplified methods have been developed to assess SOC sequestration at both regional and continental scales. For example, based on long-term experiments, statistical relationships have been used to explore regional SOC sequestration potential in European (Smith et al. 1997, 1998) and Chinese croplands (Qin and Huang 2010, Sun et al. 2010). In the United States, Sperow et al. (2003) also applied a modified version of the Intergovernmental Panel on Climate Change (IPCC) inventory approach to assess SOC in agricultural soils. However, accounting for the effects of heterogeneity in soil and climate conditions and management practices across space and time on SOC is difficult using these approaches (Falloo et al. 2002a, b).

Meta-modeling is a sophisticated procedure to derive simple relationships from process-based modeling (Kleijn and Standridge 1992). Meta-models are functions linking the output to primary drivers with calibrated parameters. They can be used as abstractions and simplifications of the simulation model and have been frequently applied in simulation studies (Kleijn and Sargent 2000, Kleijn et al. 2005). Based on simulations of the Agricultural Production Systems sIMulator (APSIM), for example, Florin et al. (2011) derived an inverse meta-model using crop yield data to estimate soil available water capacity at high spatial resolution. They found that the meta-model could explain ~70% of the variation in crop yield predicted by APSIM. There is potential for meta-modeling to facilitate regional or continental scale predictions of SOC changes without the need for detailed information to initialize process-based models.

In this study, we used the widely tested and verified process-based APSIM (Wang et al. 2002, Keating et al. 2003) to conduct simulations on SOC responses to spatiotemporal variations of climate and soil at 613

reference sites (with detailed soil parameters) located in Australia's cereal-growing regions. One management scenario of optimal fertilizer application and 100% residue return for a wheat crop was adopted in the modeling, representing the typical cropping type and the best CAP. Based on the simulated responses, we aimed to: (1) identify the most significant environmental controls on SOC sequestration potential (i.e., total SOC content at equilibrium state under optimal management) and derive a simple meta-model for SOC sequestration potential, (2) implement the meta-model at regional scale to assess the potential SOC sequestration under our defined management scenario and its spatial pattern in Australia's cereal-growing regions, and (3) quantify the uncertainty of SOC sequestration induced by variation in soil properties in Australia's cereal-growing regions.

METHODS

Soil profile and climate data for APSIM simulations

In total, profile data for 613 sites (soils) were collected from the Agricultural Production Systems Research Unit (APSRU) reference sites (*available online*),⁶ which includes the data needed to initialize soil water, carbon, and nutrient parameters in APSIM (Table 1). The 613 sites were located in cultivated lands distributed across the Australian cereal-growing regions, and covered a wide range of climatic and soil conditions (Fig. 1). Daily weather data (including rainfall, maximum and minimum temperature, radiation) at the nearest climate station to each of the 613 sites were obtained from the SILO Patched Point Dataset (*available online*).⁷

APSIM and the scenario simulations

APSIM version 7.3 was used for the simulation. APSIM was developed to simulate biophysical process

⁶ <http://www.asris.csiro.au/themes/model.html#>

⁷ <http://www.longpaddock.qld.gov.au/silo/>

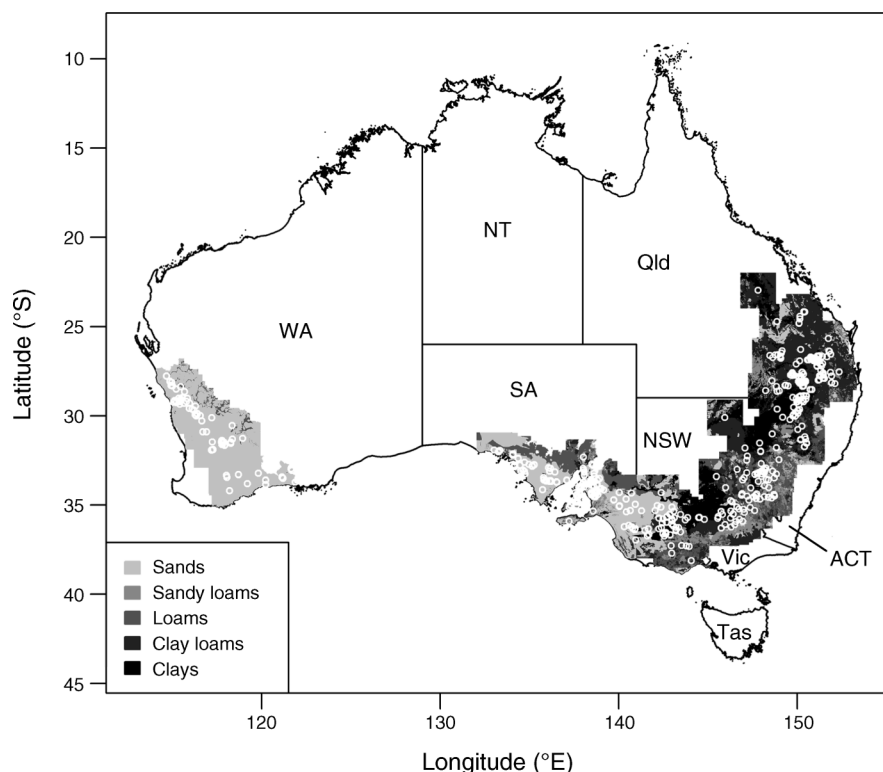


FIG. 1. The study region, soil texture, and location of the studied 613 soils (white circles) in Australia. Key to abbreviations: ACT, Australian Capital Territory; Qld, Queensland; NSW, New South Wales; NT, Northern Territory; SA, South Australia; Tas, Tasmania; Vic, Victoria; WA, Western Australia.

in farming systems, and has been comprehensively verified and used, particularly in Australia and China, to study productivity, nutrient cycling, and environmental impacts of farming systems as influenced by climate variability and management interventions (van Ittersum et al. 2003, Luo et al. 2005, Probert et al. 2005, Bryan et al. 2010, Chen et al. 2010, Wang et al. 2010). APSIM simulates crop growth and soil processes on a daily time step in response to climate (i.e., temperature, rainfall, and radiation), soil water availability, and soil nutrient status (i.e., N and phosphorus; Wang et al. 2003). It requires detailed information about management practices and soil profile characteristics as listed in Table 1. The SoilN module in APSIM simulates the dynamics of both C and N in each soil layer. SOC is divided into four conceptual pools and the decomposition of each pool is treated as a first-order decay process, leading to the release of CO_2 to the atmosphere and transfer of the remaining decomposed C to other pools. The flow of N depends on the C:N ratio of the receiving pool. Decomposition of surface residue is simulated by the SurfaceOM module, taking into account the degree of contact of residue with soil to modify the maximum potential decomposition rate of residue. APSIM allows flexible specification of management options like crop and rotation type, tillage, residue management, fertilization, and irrigation. A number of studies have

concluded that APSIM is able to predict SOC dynamics under different cropping and management systems with a high level of accuracy (Probert et al. 2005, Huth et al. 2010, Luo et al. 2011).

Wheat is the most common crop grown in Australia. The ability of APSIM to accurately simulate wheat crop growth has been widely tested and verified across Australia (Meinke et al. 1997, Asseng et al. 1998, Wang et al. 2003). We assessed the potential of C sequestration under a typical annual wheat system with optimal application of fertilizers (no nutrient deficiency) and 100% residue retention. A wheat crop was assumed to be sown every year depending on rainfall and soil water content, which varies for different regions. Wheat cultivar was assigned according to sowing date, the earlier the sowing date, the later the maturity type of the wheat cultivar. Crop residues (stem plus leaf) after harvest were retained in the system, and we assumed 10% was left on the soil surface, with the remainder incorporated into the top 20 cm of soil.

The APSIM model was run for 1022 years for each soil/site with the continuous wheat system to generate the SOC changes in each soil layer year by year. The climate data from 1889 to 2010 (the period of climate record) were used for the first 122-year simulations, and the data from 1911 to 2010 were repeated nine times for the following 900-year simulations. The simulated SOC

contents in the top 30 cm soil profile in each year, together with a list of selected variables as described in the next section, were used to derive the meta-model.

Calculations of SOC at equilibrium state

The SOC content would gradually reach the equilibrium state during the 1022-year simulation (Stewart et al. 2007, West and Six 2007) considering that the turnover time of SOC in the top soils to reach equilibrium state is usually less than a century in Australia (Barrett 2002). In our simulated annual wheat systems, we specified that the equilibrium state only fluctuates with interannual changes in local climate. To assess whether the SOC content reached the equilibrium state at the end of the 1022-year simulation, we calculated the average of SOC content in the last 100-year simulations (C_E). Then, the amount of SOC content in each year during the simulation (C_i , $i = 1, 2, 3, \dots, 1022$) relative to C_E was calculated as C_r :

$$C_r = \frac{C_i}{C_E} \times 100\%. \quad (1)$$

If C_r is greater than 100%, SOC experiences a decrease; if C_r is smaller than 100%, SOC experiences an increase; if C_r is equal to 100%, SOC experiences no change. If C_r fluctuated around 100% of the SOC content without significant deviation, it was considered to reach the equilibrium state. The moving average (MA) of continuous 10 C_r was calculated to roughly derive the years needed to reach the equilibrium state:

$$MA_n = \frac{C_r^n + C_r^{n+1} + \dots + C_r^{n+9}}{10} \quad (2)$$

$n = 1, 2, 3, \dots, 1013$

where C_r^n is the C_r after n years of simulation. When MA_n reaches the value between 0.99 and 1.01, the SOC content was considered to reach the equilibrium state after n years. The absolute change in SOC (C_c) after 1022 years of simulation was also calculated as the difference between modeled C_E and the recorded initial SOC (C_0), i.e., $C_c = C_E - C_0$.

Rationale of meta-modeling

The rationale for meta-modeling is to derive simplified relationships, called meta-models, from process-based modeling (Kleijn and Standridge 1992). Meta-models capture the impact of the primary drivers on the output of interests, without the need for detailed input information that is normally not available across a wide region. For instance, the SOC change in agricultural land is primarily driven by carbon input determined by crop biomass production and carbon output determined by SOC decomposition. In semiarid regions like Australia's cereal-growing regions, crop biomass production is mainly limited by the availability of water and nutrients, and SOC decomposition by soil moisture and temperature. Other variables have only marginal effects

on the carbon input and output terms, and inaccurate estimation of those variables can also lead to increased uncertainty in the prediction (Falloon and Smith 2003, Ogle et al. 2007, 2010). By expressing SOC balance as a simplified function of the primary and surrogate drivers that are available across regions (i.e., a meta-model), prediction of SOC dynamics could be made at regional scales to support the assessment of SOC sequestration potential.

Based on the input and output of the APSIM simulations, the relationship between the output (say) w and the inputs $v_1, v_2, \dots$, can be represented through the function f_0 (Kleijn and Standridge 1992):

$$w = f_0(v_1, v_2, \dots). \quad (3)$$

An increased number of input variables (v) would need to be incorporated into f_0 to get the optimal estimation of the output (w). We attempted to derive a meta-model that described the relationship f_0 through a few input variables and parameters that have significant impact on SOC decomposition (C output) and crop production (C input), including: (1) initial active and/or decomposable SOC content (C_a), inert and/or recalcitrant organic carbon content (IOC), initial fraction of the microbial biomass C pool (F_{BIOM}) and IOC pools (F_{IOC}) in total soil organic carbon, soil C:N ratio (CNR), soil pH (pH), plant available water capacity of soil (PAWC) (all of these variables were weighted averages in the top 30 cm of soil except for PAWC, which was calculated in the whole soil profile); and (2) total radiation (R), precipitation (P), and mean temperature (T) during the wheat growing season (i.e., from 1 May to 31 October).

Development of the meta-model

The meta-model can be estimated using Taylor series approximation, which enables the approximation of f_0 with a polynomial. As inert organic carbon (IOC) is not subject to change during the simulation, we focused on the total active SOC content, C_a (i.e., total SOC excluding IOC) for the meta-model. The IOC content will be added to the model as a random effect when applying the model at regional scale. We started with a first-order polynomial:

$$y = \beta_0 + \sum_{k=1}^n \beta_k x_k + \varepsilon \quad \varepsilon \sim \mathcal{N}(0, \sigma^2) \quad (4)$$

where y is the modeled C_a at equilibrium state, x_k the possible k covariates based on the model inputs (e.g., C_a , F_{BIOM} , F_{IOC} , CNR, pH, PAWC, P , R , and T), β the regression coefficients, i.e., the parameters need to be estimated, and ε the residual error, which is normally distributed with a mean 0 and standard deviation σ . The σ represents the variability with which the output deviates from their predictions based on the meta-model. Stepwise multi-regression (forward and backward) was applied to find the best fitted meta-model based on Akaike's Information Criteria (AIC).

We further tested higher-order polynomial meta-models including all possible interactions and quadratic effects. The leave-one-out cross-validation procedure was applied to test the validity of the meta-model (Kleijnen and Standridge 1992). After deriving the meta-model, partial regression was also performed to assess the effect of each selected variable on predicted C_a .

Soil and climate layers for meta-modeling at regional scale

Texture is a basic property of a soil with a significant effect on soil water, porosity, nutrient cycling, and microbial reactions, and is not readily subject to change (Brady and Weil 2008). Based on soil texture of the topsoil (A horizon), we divided 597 soils excluding 16 soils without soil texture information into five groups (Fig. 1): sands (250 soils), sandy loams (23 soils), loams (66 soils), clay loams (116 soils), and clays (142 soils). For each soil-related covariate (included the derived meta-model) in each soil group, its probability density function (PDF) was constructed using the data of the 597 reference soils. First, we selected a PDF family from several hypothesized families of distributions including normal, lognormal, gamma, Weibull, and uniform distributions by calculating the skewness and kurtosis of the covariate. Then, the parameters of the PDF for each covariate in each soil group were determined by maximum likelihood estimation. The PDFs of inert organic carbon (IOC), initial active organic carbon (C_a), the fraction of initial C_a in initial total SOC (f_{Ca}), and initial total SOC (C_0) also were identified following the same process in order to estimate total SOC and its change.

To define the study area, we centered a $2^\circ \times 2^\circ$ grid cell on each of the 597 reference sites and created a continuous region by overlapping the grid cells (Fig. 1). Soil texture data across the whole region were collected from the database of the Australian Soil Resources Information System (ASRIS) (*available online*; Fig. 1).⁸ The soil texture layer of ASRIS has a resolution of $0.01^\circ \times 0.01^\circ$ (latitude $\times$ longitude) and was estimated using classification models based on 993 165 texture observations (Henderson et al. 2001). Climate data across the whole region at a spatial resolution of $0.05^\circ \times 0.05^\circ$ (totaling 8165 points across the region) were collected from the Bureau of Meteorology, Australia (Jeffrey et al. 2001; *available online*).⁹ The data span the years 1889–2010 on a daily time step. The nearest climate data point to the pixel was used to calculate the climate-related covariates in the derived meta-model.

Regional application and uncertainty analysis of the meta-model

For each pixel ($0.01^\circ \times 0.01^\circ$) of the soil texture layer, the derived meta-model was applied to calculate active

SOC (C_a) content based on the corresponding soil- and climate-related covariates. A Monte Carlo analysis was used in each pixel to address uncertainty induced by soil-related covariates (i.e., pH and PAWC, see details in *Results*). First, 2000 series of values were sampled from PDFs for each of the soil-related covariates and the residual error ε assuming normality. Then, 2000 estimates were computed using the derived meta-model based on the 2000 series of values of the soil-related covariates and ε . Other covariates (i.e., P , T , and R) used the constant identified in each pixel based on the 8165 climate data points. Here, we assumed that active organic carbon (C_a) was not negative. If C_a calculated by the derived meta-model was negative, a new series of soil-related covariates was resampled to ensure that C_a was not negative. Inert organic carbon (IOC) was added to the 2000 estimates of C_a as a random effect to get the final total SOC by sampling 2000 replicates from the PDF of IOC for each soil type. Finally, the 2000 estimates of the total SOC ($C_a + \text{IOC}$) were used to construct the 95% confidence intervals (CI) to indicate the uncertainty of total SOC. A percentage uncertainty (PU) was estimated based on half of the 95% CI divided by the average (Ogle et al. 2010):

$$\text{PU} = \frac{\text{CI}_u - \text{CI}_l}{2 \times m} \times 100\% \quad (5)$$

where CI_u and CI_l are the upper and lower limits of the 95% CI, respectively; m is the average of the 2000 estimates. The uncertainty of total SOC included two components. One was induced by the soil-related covariates identified by the meta-model, and the other was induced by the random effect of IOC. The uncertainty of IOC was also calculated by constructing the 95% CI of the above 2000 estimates of IOC. The PU induced by IOC (PU_{IOC}) was estimated using the CI_u and CI_l of IOC according to Eq. 5. Then, the ratio of PU_{IOC} to PU was calculated to indicate the fraction of the uncertainty of SOC resulting from the uncertainty of IOC.

A 95% CI of SOC change (i.e., C_c) was also estimated based on the 2000 replicates in each pixel. To calculate the SOC change, the initial SOC content in each pixel was estimated by adding the estimates of IOC and estimates of initial C_a . For each estimate in each pixel, initial C_a was conditionally sampled from PDF of C_a depending on IOC and f_{Ca} to ensure that the sum of C_a and IOC is not beyond the range of the 2000 estimates of C_0 , and the fraction of the sampled C_a not beyond the range of the 2000 estimates of f_{Ca} . All the analyses were conducted using R 2.13.1 (R Development Core Team 2011).

RESULTS

APSIM simulations and the meta-model

During the 1022-year simulation of APSIM, SOC gradually reached equilibrium (Fig. 2). Among the 613 sites, the time needed to reach the equilibrium ranged from 5 to 922 years. Multi-regression analysis indicated that the initial amount of active C (i.e., C_a), precipita-

⁸ <http://www.asris.csiro.au/>

⁹ <http://www.bom.gov.au>

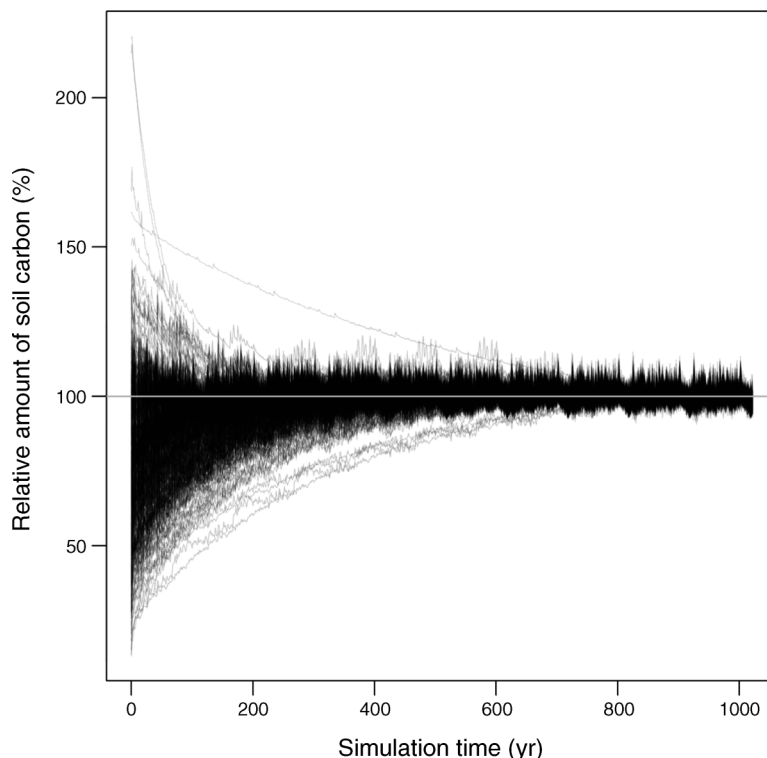


FIG. 2. The relative change of soil organic carbon (C_t) in the top 30 cm soil in each year to the average of SOC content in the last 100-year simulation (C_E) based on APSIM simulations in 613 soils in Australia's cereal-growing regions.

tion (P), plant available water capacity of soil (PAWC), and temperature (T) had significant effect on the time needed to reach the new equilibrium state, and can explain 51% of the variation (data not shown).

At the end of the simulation, SOC increased at 547 sites and decreased at 66 sites with an average across all sites being an increase of 18.40 Mg/ha. On average, the baseline SOC content was 67.32% of the SOC content at the end of the 1022-year simulation under the continuous wheat system (Fig. 2).

At the 613 reference sites, precipitation (P) during the wheat growth season (from 1 May to 31 October) was positively and significantly related to modeled final active organic carbon (C_a) by APSIM (Table 2), whereas the effects of soil pH, PAWC, temperature (T), and radiation (R) during the wheat growth season were negative and significant (Table 2). Overall, the modeled C_a content could be meta-modeled by a first-order polynomial with a prediction error ε , which has independent normal distributions with mean 0 and stand deviation 5.96 Mg/ha (Table 2):

$$C_a = 95.18 + 0.071P - 0.017PAWC - 1.94pH - 0.016R - 1.42T + \varepsilon. \quad (6)$$

This first-order meta-model had an AIC of 2193.54 and explained 74% of the variation in C_a as modeled by APSIM (Fig. 3), but required much fewer input

variables (that are easily obtainable) than APSIM (Table 1).

Results from the partial regression (Fig. 4) indicate that precipitation had the highest impact on final active organic carbon (C_a), and explained 24% of the variation in modeled C_a (Fig. 4A). Radiation and soil pH had moderate negative effect on modeled C_a , respectively (Fig. 4B, C). Temperature and PAWC had the least effect (Fig. 4D, E), although their effects were significant.

The higher-order polynomial was also fitted by considering the interactions and quadratic effects of

TABLE 2. Coefficients of stepwise multiple regression analysis of the effect of soil pH, plant available water capacity of soil (PAWC), mean annual precipitation (P), mean annual radiation (R), and mean annual temperature (T) on soil carbon content at the end of the 1022-year simulation.

Effect	Soil carbon content		
	Estimate	<i>t</i>	<i>P</i>
Intercept	95.18	22.22	<0.0001
pH	-1.94	-8.23	<0.0001
PAWC	-0.017	-4.29	<0.0001
<i>P</i>	0.071	13.89	<0.0001
<i>R</i>	-0.016	-9.8	<0.0001
<i>T</i>	-1.42	-5.49	<0.0001

Note: Whole-model $R^2 = 0.74$, $P < 0.0001$.

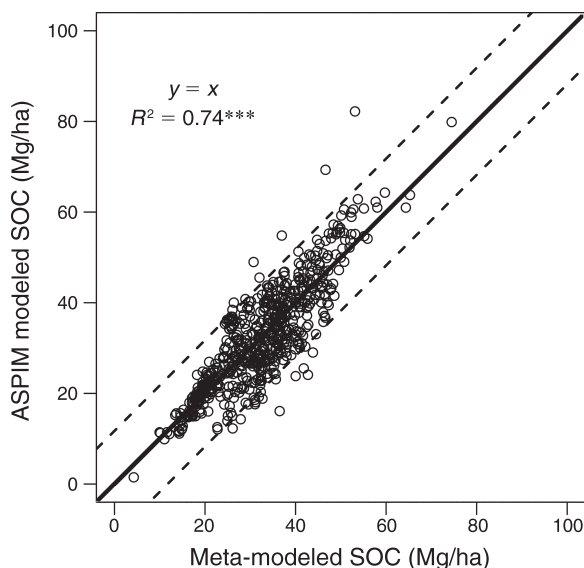


FIG. 3. The relationship between meta-modeled and Agricultural Production Systems sIMulator (APSIM) modeled soil organic carbon (SOC) content in the top 30 cm soil at the end of the 1022-year simulation for the 613 soils based on the leave-one-out cross-validation. The black dashed lines show the 95% meta-modeled interval. The solid black line shows the 1:1 line. *** $P < 0.0001$.

the possible covariates (data not shown). The best model performed slightly better than the first-order model, explaining 79% of the variation of APSIM-modeled C_a with an AIC of 2089.77. Due to the much easier form and interpretability, we selected the first-order meta-model (Eq. 6) coupling with the random effect term of IOC to estimate total SOC sequestration potential in Australia's cereal-growing regions by considering the uncertainty of soil-related covariates, i.e., pH and PAWC.

Regional SOC results from meta-modeling

The meta-model-predicted SOC sequestration potential at equilibrium in all pixels ranged from 21.14 to 152.71 Mg/ha with a mean of 52.11 Mg/ha across the study area (Fig. 5). Predicted change in SOC (C_c) ranged from -25.01 to 107.57 Mg/ha with a mean increase of 8.17 Mg/ha (Fig. 5).

Under the assumed continuous wheat system, average SOC content exhibited regional patterns (Fig. 6). Generally, it decreased from southwest to northeast in Western Australia; and decreased from southeast to northwest in other parts of the study area (Fig. 6B). The upper and lower limit of the 95% confidence interval showed similar regional patterns to the average SOC (Fig. 6A, C). In the regions with lower SOC content, however, there was greater uncertainty induced by variation in soil properties. This resulted in increasing uncertainty from southwest to northeast in Western Australia, and from southeast to northwest in the other parts of the study area (Fig. 7A). The average

percentage uncertainty (PU) was 44% ranging from 12% to 117% in the whole study area (Fig. 7A). Among the sources of uncertainty (soil pH, PAWC, IOC), on average, IOC accounted for 73% of the average percentage uncertainty ranging from 50.44% to 99.81% without apparent spatial pattern (Figs. 1 and 7B).

The absolute change of SOC also showed regional patterns (Fig. 8). Average of the absolute change in SOC (C_c) varied from a large increase (>100 Mg/ha) in the southwest of Western Australia and the southeast of the rest of the study region to a moderate decrease (less than -20 Mg/ha) in the northeast of Western Australia and northwest of the rest of the study region (Fig. 8B). The lower and upper limit of the 95% CI showed similar regional patterns to the average C_c (Fig. 8A, C). In most of the study area, on average, absolute SOC change ranged from -15 to 30 Mg/ha. The greatest increases of >100 Mg/ha occurred on the edge of the study area near the border between New South Wales and Victoria (Fig. 8B). In the northern-most parts of Queensland and Western Australia, soil carbon change (C_c) could be either negative or positive depending on soil properties (Fig. 8A, B). On the edge of the study area near the border between New South Wales and Victoria, soil carbon change was always a C sink regardless of soil properties (Fig. 8A, B).

DISCUSSION

Credibility and benefits of meta-modeling

The results presented here indicate that soil C sequestration potential (i.e., the SOC content at equilibrium under optimal management) simulated by detailed process-based models could be adequately captured by meta-modeling using just a few, easily obtainable spatial soil and climate parameters. Although our simulation was not based on observed data, the results show that it is feasible to simplify the modeling and capture the dynamics of SOC across different regions by considering the underlying primary controls. Our methods build upon previous simplified methods that model agricultural soil C sequestration at a regional scale (Smith et al. 1997, 1998, Sperow et al. 2003, Qin and Huang 2010, Sun et al. 2010), and provide a way to include spatial details and impact of drivers such as soil pH, soil water holding capacity, and climate variables. The meta-modeling approach avoids the need to initialize and parameterize process-based models with detailed information, and can be readily applied to regional scales at high resolution.

Similar efforts to summarize the impact of primary drivers on SOC were also made in previous studies. Based on data from 11 temperate and tropical grassland sites, Parton et al. (1993) found that the Century model and an empirical regression model derived based on climate variables showed similar performance in predicting aboveground biomass and soil C content. Based on simulation results of the Century model, Schimel et al. (1994) summarized the effects of temperature and soil

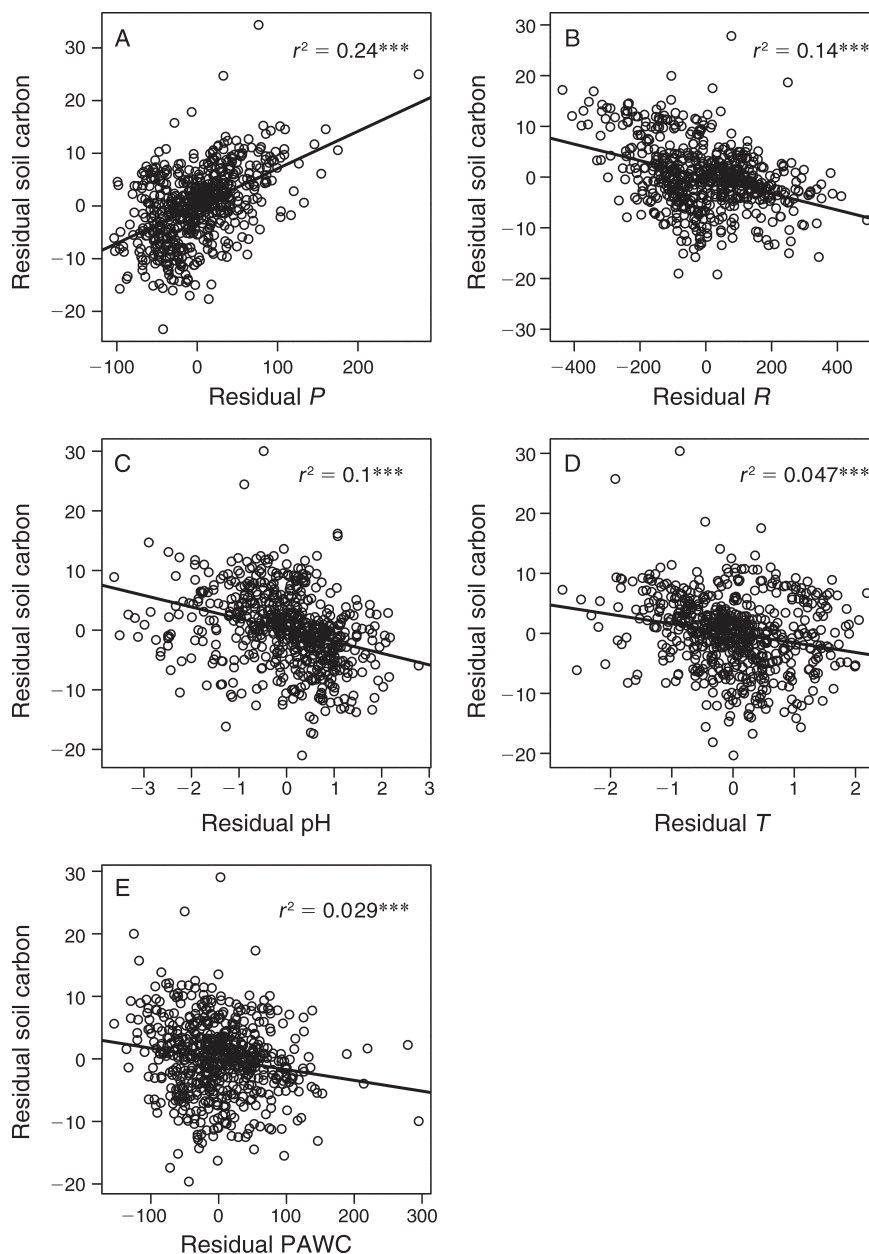


FIG. 4. Partial regression for the multiple regression model, i.e., meta-model, presented in Table 2. Partial regression plots are shown for residuals of (A) mean annual precipitation (P , measured in mm), (B) mean annual radiation (R , measured in MJ/m^2), (C) soil pH, (D) mean annual temperature (T , measured in $^{\circ}\text{C}$), and (E) plant available water capacity of soil (PAWC, measured in mm), on residuals of soil carbon (measured in Mg/ha), while controlling for all other identified variables for the meta-model.

*** $P < 0.001$.

texture on SOC and its turnover time using simple statistical equations, and used the simple models to estimate global SOC turnover time. The newly developed meta-model in this paper adds to these previous efforts, providing a simplified approach to estimate SOC change and sequestration potential under cereal cropping and best practices as affected by climate and soil variations across a region of Mediterranean to semi-arid climate. However, when employing the derived meta-

model in this study in other regions such as subtropical and tropical agroecosystems, generality of the model needs to be evaluated (Jung et al. 2009).

Controls on SOC content identified by meta-modeling

Three climatic variables, i.e., long-term mean temperature (T), precipitation (P), and radiation (R) during the crop growth season, together with the plant available water capacity of soil (PAWC), had significant effects on

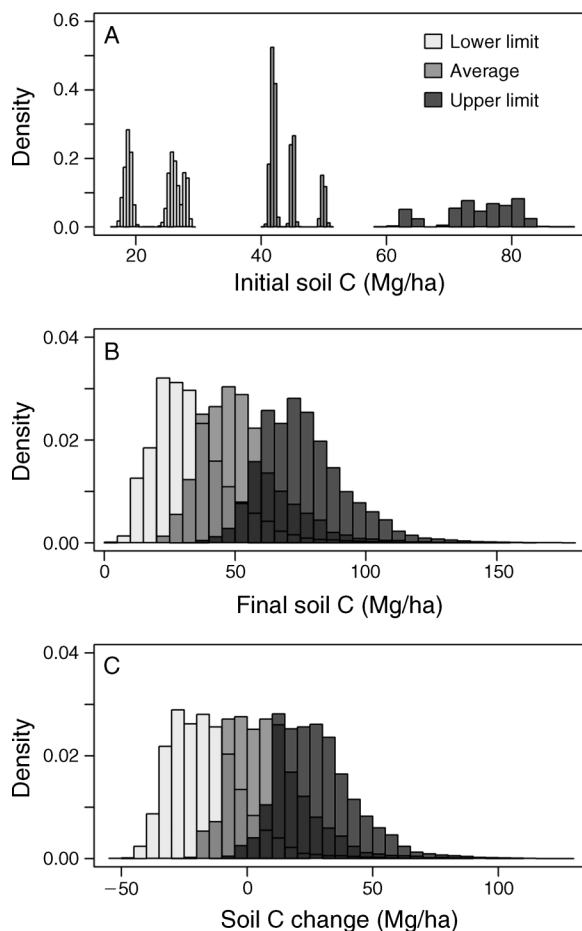


FIG. 5. The probability density distribution of (A) the initial soil organic carbon content, (B) final soil organic carbon content at equilibrium state modeled by the meta-model, and (C) their difference, i.e., the soil organic carbon change in all pixels ($0.01^\circ \times 0.01^\circ$) in Australia's cereal-growing regions. Light and dark gray bars show the lower and upper limit of the 95% confidence interval, respectively. Medium gray bars show the average.

SOC. In drier climates, crop productivity (SOC input) is mainly limited by water availability as determined by water supply (P) and storage (PAWC). Both temperature and moisture conditions in soil also affect soil respiration through decomposition (SOC output; Davidson and Janssens 2006, Mahecha et al. 2010). The positive coefficients for precipitation (P) and the negative coefficients for PAWC, temperature (T), and radiation (R) on SOC in our meta-model indicate that the improvement in water supply reflected by increased values for P , leads to a SOC increase. Conversely, increase in temperature and radiation increases SOC decomposition and reduces water availability (with high R leading to high evapotranspiration), resulting in a decrease in SOC. The significant negative effect of PAWC is more difficult to explain. Increases in PAWC improve soil moisture conditions, which would not only benefit crop growth and potential soil C input but also

accelerate soil C decomposition. The relative importance of the two consequences in different climatic regions needed to be further studied.

The meta-modeling indicated that soil pH had significant effects on predicted SOC, and the lower the soil pH, the higher the SOC stocks. This result can be attributed to the inhibited microbial activity in lower soil pH and is consistent with previous studies on the effects of soil pH on SOC dynamics (Motavalli et al. 1995, Benbi and Brar 2009).

Limitations and uncertainties of the application of the meta-model in this study

Variation in agricultural management (which we did not consider) could influence SOC dynamics. First, we assumed a continuous wheat system under optimal agricultural practices with no nutrient deficiency and 100% residue retention across Australia's cereal-growing regions. Although wheat production is common in the region, these assumptions can under- or overestimate soil C sequestration potential depending on cropping systems. For example, alternative fertilization practices, such as using manures and other organic fertilizers, and increased rotation complexity (e.g., increasing crop diversity and introducing legumes into rotation) may significantly enhance SOC accumulation compared with single crop systems (West and Post 2002, Luo et al. 2010a). Second, as the APSIM model cannot fully capture the effects of different tillage types on soil environments and their subsequent influences on SOC, impact of tillage (such as no-till) on SOC change was not included in the current study. However, in terms of long-term change in total SOC, such impact may be less significant because tillage was found mainly to redistribute SOC in the soil layer down to tillage depth but not significantly change the total SOC content (Luo et al. 2010b). Third, nitrogen is a dominant limiting factor for crop growth in many regions. The assumption of no nitrogen limitation would overestimate the estimation of SOC sequestration potential. To minimize risk under real world conditions (e.g., the cost of fertilizers), in fact, the actual rates of nitrogen applied are likely to be lower than the optimal nitrogen rate for eliminating soil nitrogen deficiency (Wang et al. 2008) as simulated in this study. Fourth, there is a lot of remnant vegetation in the study area where the land has never been cropped. In these uncultivated ecosystems, the SOC dynamics would be generally at equilibrium state. To reduce the uncertainty resulting from the diverse cropping systems and agricultural practices, more detailed simulation and testing of both process-based models and the corresponding meta-modeling are needed based on detailed land use and management information.

Potential of SOC sequestration in Australia's cereal-growing regions

Our results suggest that soils in Australia's cereal-growing regions may be able to sequester carbon under a

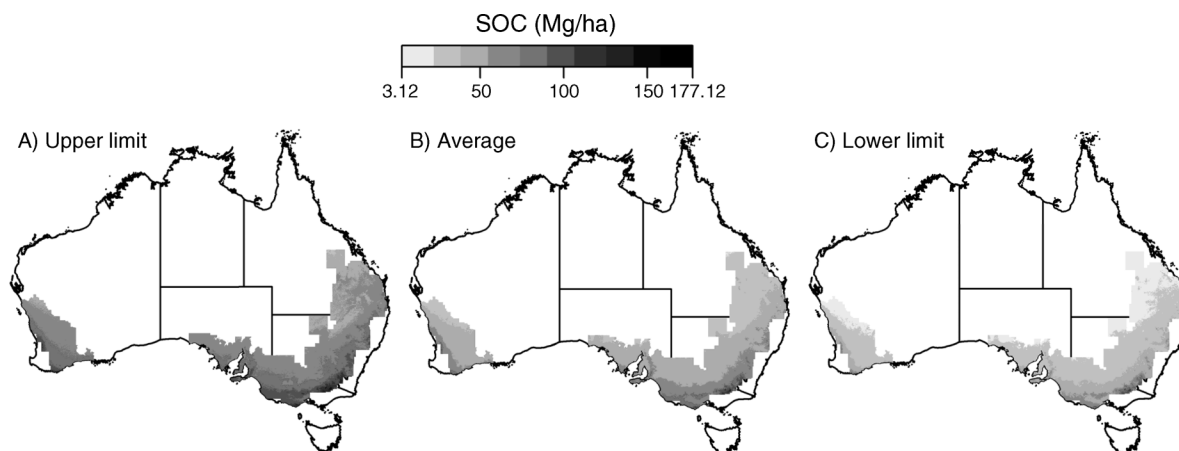


FIG. 6. The meta-model modeled regional pattern of soil organic carbon content (Mg/ha) at the end of simulation under continuous wheat system in Australia's cereal-growing regions. Panels A, B, and C show the upper, average, and lower limit of the 95% confidence interval as induced by the availability of soil properties, respectively. The smallest and highest values are shown at the start and end of the legend bar, respectively.

typical annual wheat system if best management practices could be adopted, i.e., optimal application of fertilizers and 100% residue retention. On average, the modeled total potential SOC stock in the top 30 cm soil is ~ 7.5 Pg calculated by summing the average SOC stock in each pixel in the studied region. Its 95% confidence interval was 4.8–10.9 Pg. Compared with initial SOC stock, SOC stock increased by ~ 1.2 Pg under our annual wheat system scenario with a 95% confidence interval of -2.3 – 4.0 Pg across the whole study region. If we consider that the total area sown to wheat in the study area in 2011–2012 is ~ 14.3 million ha ($\sim 10\%$ of the total study area; Australian Crop Report 2011), the magnitude of the SOC increase at equilibrium under 100% residue retention and optimal fertilization in actual wheat cropland would be ~ 0.12 Pg. This amount of C sequestration potential may offset $\sim 16\%$ of the C emissions from loss of aboveground biomass from the Australian landscape (0.76 Pg with the 95% confidence interval from 0.35 – 0.38 to 1.1 – 1.3 Pg depending on scenarios) due to cropping since European colonization of Australia (Barrett et al. 2001), and is almost equal to the national greenhouse gas emission of 0.56 Pg CO_2 -equivalent (i.e., 0.15 Pg C) in 2009 (available online).¹⁰ The economic viability of the best practice simulated in this study needs to be further investigated.

The results are comparable with those from other studies in Australian croplands. For instance, based on a simple reversed calculation by comparing SOC content in uncultivated soils with that in cultivated soils, Dalal and Chan (2001) suggested that the potential sinks would be 1.04 Pg CO_2 -equivalent if adopting conservation agriculture practices such as residue retention for 20 years in Australia's cereal-growing regions. Through an analysis of the published evidence for SOC stock

changes resulting from shifts in agricultural practices, Sanderman et al. (2010) also reported that improved management of cropland, such as no-tillage and residue retention, has resulted in a relative gain of 0.2 – 0.3 Mg $\text{C}\cdot\text{ha}^{-1}\cdot\text{yr}^{-1}$ compared to conventional agricultural practices across a range of Australian soils. If we extrapolate their estimate to our study area, the total potential is ~ 0.58 – 0.87 Pg over 20 years. However, those estimates ignored the fact that SOC sequestration could continue for 50–150 years after adopting conservation agricultural practices (West and Post 2002, West and Six 2007, Qin and Huang 2010). This explains why the SOC sequestration potential estimates of Sanderman et al. (2010) are lower than our meta-modeled results.

Uncertainties of SOC sequestration potential induced by variability of soil properties

Limited soil information is a significant source of uncertainty at regional or continental scales. By introducing variation in soil parameters (pH and PAWC) and adding the uncertainty of inert organic carbon (IOC) into the meta-model, the results showed significant spatial variability of the uncertainty of SOC sequestration potential with apparent spatial pattern relating to temperature and rainfall regimes across the study region. Generally, the variation of soil properties led to smaller uncertainty in cool and wet environments than in warm and/or dry environments. Wynn et al. (2006) also found that the effects of soil texture on SOC stabilization is diminished in cool, temperate environments, where other factors may play a more significant role in inhibiting SOC decomposition, particularly temperature control on respiration rates. This phenomenon could result from water and/or temperature limitations on soil C input and/or output across eco-climatic zones (Wynn et al. 2006, Wang et al. 2009). For example, with similar rainfall, the ability of soils to store

¹⁰ <http://ageis.climatechange.gov.au/>

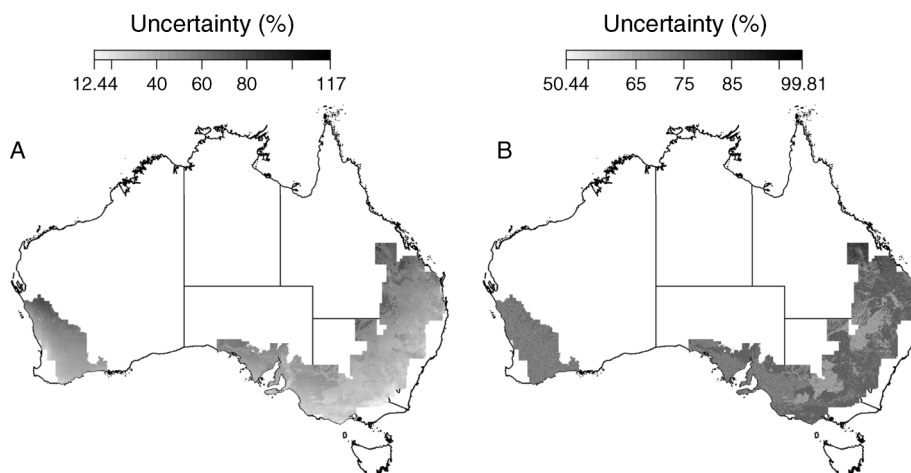


FIG. 7. (A) The meta-model modeled percentage uncertainty of total soil organic carbon content (%) and (B) the fraction of the uncertainty attributable to uncertainty of inert organic carbon (%) at the end of simulation under continuous wheat system in Australia's cereal-growing regions. The smallest and highest values are shown at the start and end of the legend bar, respectively.

soil moisture would be more important in warm regions compared to cool regions because of the higher chance of water limitation in warm regions. Moisture storage would also be more important in dry regions compared to wet regions with similar temperature. Thus, variation in soil properties would contribute to higher diversity of soil moisture in warm regions, thereby leading to higher variability of both soil C inputs by crop production and outputs by decomposition, and then the higher uncertainty of final SOC content. These results suggested that soil conditions would be more important for regulating SOC sequestration in warm and/or dry environments than that in cool and/or wet environments. In warm and/or dry environments, the uncertainty induced by the variability of soil conditions should be accounted for when assessing SOC sequestration potential and advising agricultural practices, particularly when upscaling

local scale results to regional and continental scales. However, more detailed studies are needed to explicitly address the interacting effects of both climate and soil conditions on SOC input and output. This is important for improved estimation of SOC sequestration and for understanding the mechanisms controlling SOC dynamics at regional or continental scales.

CONCLUSIONS

Our simulations demonstrate the ability of meta-modeling to summarize complicated process-based models and predict SOC dynamics in agroecosystems. The meta-model identified the underlying covariates of rainfall, temperature, and soil water holding capacity controlling SOC dynamics, and significantly reduced the demand for input data required by process models. The meta-modeling results indicated that Australia's cereal-

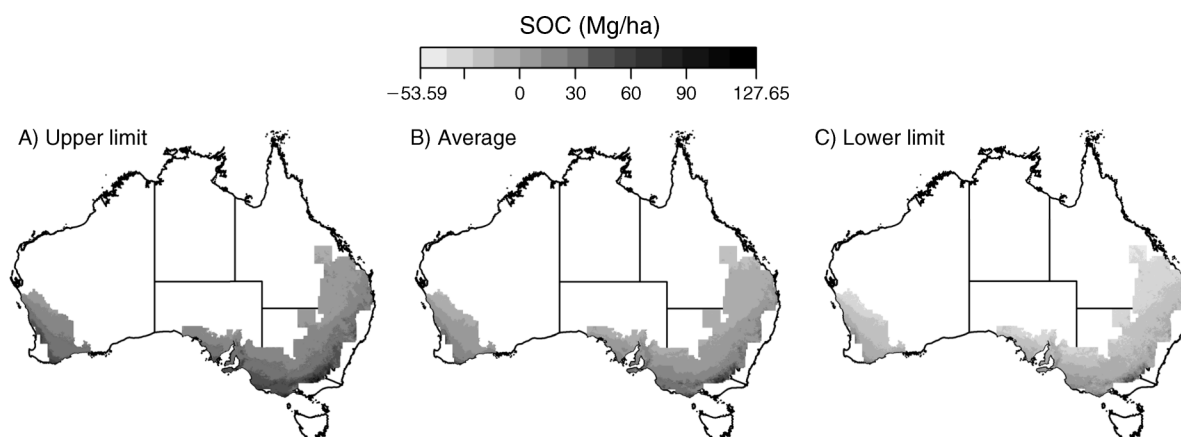


FIG. 8. The meta-model modeled regional pattern of soil organic carbon change (Mg/ha, C_e) at the end of simulation under continuous wheat system in Australia's cereal-growing regions. Panels A, B, and C show the upper, average, and lower limit of the 95% confidence interval as induced by variability of soil properties, respectively. The smallest and highest values are shown at the start and end of the legend bar, respectively.

growing regions are a C sink with a SOC sequestration potential of ~ 1.2 Pg (~ 0.12 Pg in wheat croplands) under a continuous wheat system with no nitrogen deficiency and 100% residue retention. However, variation in soil properties, particularly the amount of initial SOC including inert and active organic carbon, can result in significant uncertainty of SOC sequestration potential depending on climate, with higher uncertainty in warm, wet regions. More comprehensive understanding of the interaction of soil and climate conditions and more detailed information on land use and agricultural management are required for precise regional estimation of SOC sequestration.

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